

Lecture 35

4/20/26

JRA

Housekeeping

- Final exam will be in-class w/ a take-home portion on digital image processing
- Very similar to practice final
- Concepts highlighted in Review of Medical Imaging

MRI (cont)

Strengths

Amazing soft-tissue contrast
Anatomical & functional
Benign radio-frequency radiation
Good resolution
Phenomenal contrast

Weaknesses

Slow
Expensive

Phenomenally complicated, so simplify here

Surprise #1 - image is formed using the intrinsic magnetism of hydrogen (Hydrogen nucleus)

MRI can be based on any nucleus w/
an odd number of protons & neutrons

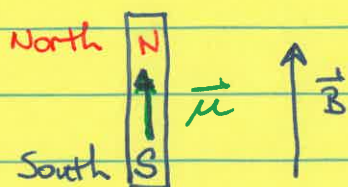
Look at hydrogen nucleus because

- it has large magnetic moment
- it's abundant

Hydrogen is the workhorse

What is the magnetism of the proton?

Proton behaves somewhat like a bar magnet
or compass needle



Compass needle aligns w/ the earth's
magnetic field

Define a magnetic moment vector $\vec{\mu}$

SIMULATION

Understanding Dipoles ...

Ran simulation, saw magnetic moment align w/ $\vec{B}$ field

Energy is lowest when $\vec{\mu}$ & $\vec{B}$ are aligned

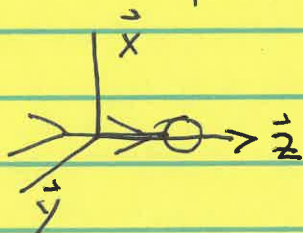
Key idea: compass needle in an external $\vec{B}$ field w.ll orient along the field (i.e., $\vec{\mu}$ w.ll align w/ $\vec{B}$) to minimize energy

Proton behaves very similarly in an external field

Protons normally randomly oriented

In MRI, doctor applies a strong $\vec{B}$ field

The $\hat{z}$ direction is taken to be horizontal (because patient is horizontal)



Protons aligned
toward $\hat{z}$

Quantum in MRI

QM says that $\vec{\mu}$ can have only two (2) values, along field direction.

This is also true for electron $\Rightarrow$ basis of electron spin resonance (ESR)

$$\vec{\mu} = \pm 1.41 \cdot 10^{-26} \text{ J/T} \quad \text{Tesla} = \text{unit of magnetic field strength}$$

Compare w/ earth's magnetic field

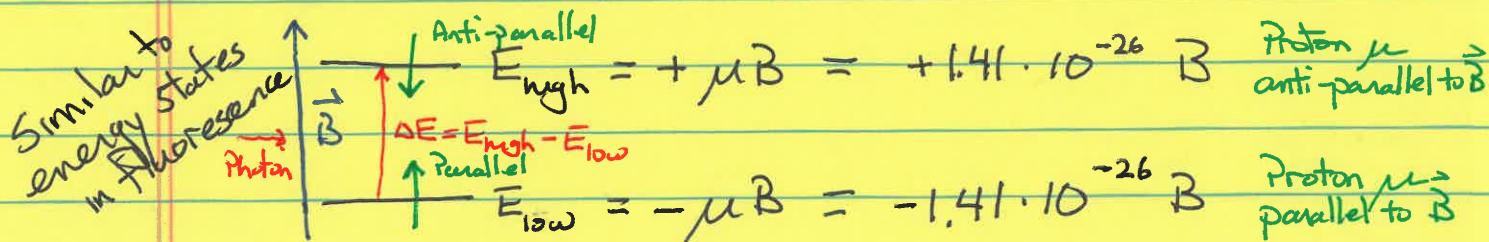
$$B_{\text{earth}} = 10^{-4} \text{ T}$$

$$B_{\text{MRI}} = 3 \text{ T} \quad \text{30,000x stronger!}$$

There are two energies associated w/ two values of μ

$$E = \pm \mu B$$

Doctor induces transitions between $E_{\text{high}} \leftrightarrow E_{\text{low}}$



This is a simplified view

The two states don't exist unless you have external field

What frequency of radiation induces transitions?

This is most imp't equation in MRI

It corresponds to being "on-resonance," w/ energy corresponding to $E_{\text{high}} - E_{\text{low}}$

★ MEMORIZE!

$$\nu_{\text{Larmor (resonance frequency)}} = \gamma B$$

$$\gamma = \text{gyromagnetic ratio of proton} \\ = 42.58 \text{ MHz/T}$$

Here is the derivation

$$E_{\text{high}} - E_{\text{low}} = h\nu_L$$

$$1.41 \cdot 10^{-26} \text{ J} - (-1.41 \cdot 10^{-26} \text{ J}) = h\nu_L$$

$$2.82 \cdot 10^{-26} \text{ J} = 6.63 \cdot 10^{-34} \nu_L$$

$$4.258 \cdot 10^7 \text{ J} \nu_L$$

$$42.58 \frac{\text{MHz}}{\text{T}} \cdot B = \nu_L$$

Key Implications of $\nu_L = 42.58 \cdot B$

(1) What kind of radiation is used?

Take $B = 2T$

$$\nu_{\text{RES}} = 42.5 \cdot B = 85 \text{ MHz}$$

This is radio frequency radiation (like FM radio)

Low energy $\Rightarrow$ NOT harmful

(2) How do you localize signals in MRI?



Which pixel generates the signal?

Aha moment: In NMR, want homogeneous field, so only chemical properties give rise to signal!

In MRI, make magnetic field vary w/ position to localize signals

$\uparrow$ Applying field gradient (i.e., variations)

SIMULATION

Investigate Frequency Encoding ...

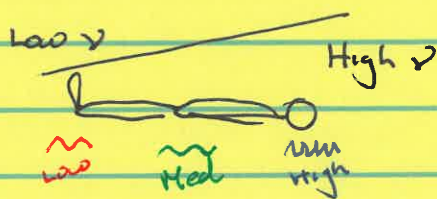
Looked first at case w/ NO gradient (i.e., constant $\vec{B}$)

This means that $\mu_{\text{RES}} = \text{constant}$ at all points in patient $\Rightarrow$ NO LOCALIZATION

Frequency is used to localize signals

- IF $B = \text{constant}$ $\nu_{\text{RES}} = \text{constant}$ NO Localization
- IF B varies w/ position ν_{RES} will vary w/ position $\Rightarrow$ Localization can be done using Frequency

SIMULATION B varies $B(z) = B_0 + G_z z$



Where B is large ν_{RES} is large

Need some way to define location

US - echo time

MRI - frequency encoding

How do you get contrast?

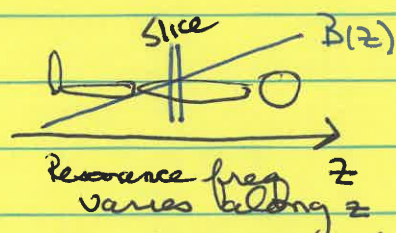
How do you frequency encode (in detail)?

We are 3 dimensional

How many gradients do you need? 3

(a) First gradient = slice selection gradient

This is conventionally taken to be along z



Apply gradient along z

Send in RF radiation pulse w/ a small range of frequencies

Only a small slice will be on resonance. Everywhere else will be too low B to resonate or too high B to resonate

This selectively excites one slice in body because every part of body has a different frequency

Doctor excites only those protons in the body whose resonance frequency ν_{RES} matches the frequency sent in

EX]

$$\nu = 100 \pm 10 \text{ Hz}$$

Ridiculous numbers
just for example



Just resonate here, from 90 to 110

This is the only place that will give a signal because it's the only place that gets excited

⇒ "Selective excitation"

Next lecture: How you measure $y, \hat{z}$

Then contrast